Statistical Thinking for Data Science

Week 3: Where the data came from

Sampling, study design, and the people you never reach

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First Semester, 2026/2027

Lecture notes for students of Statistical Thinking for Data Science.

Week 3: Where the data came from

Learning Objectives

By the end of this week you should be able to:

- distinguish the population, the sampling frame and the sample, and identify who falls between them;
- describe simple random, systematic, stratified, cluster and multistage sampling, and choose between them for a given job;
- compute a standard error and a margin of error for a proportion, and adjust for a finite population;
- compute and interpret a design effect, and say what it costs you;
- quantify non-response bias rather than apologise for it;
- tell an observational study from an experiment, and say what each can support.

3.1 Three different groups of people

Almost every mistake in this week's material comes from blurring three things.

Definition 3.1 (Population, frame, sample)**.** The **population** is everyone the question is about. The **sampling frame** is the list you actually draw from. The **sample** is who ends up in your data.

The gap between population and frame is *coverage error*, and no sample size fixes it. The gap between frame and sample is *non-response*, and it is usually worse than people admit.

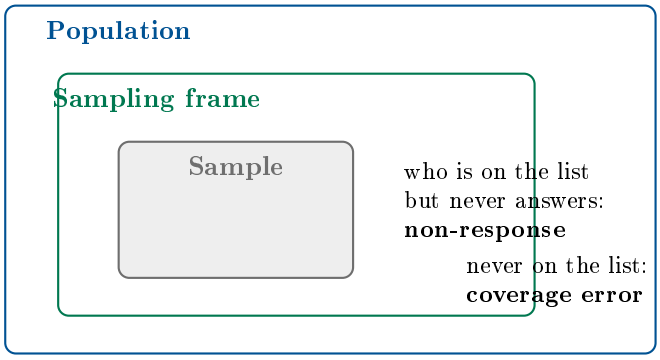


Figure 3.1: Two different gaps. Enlarging the sample closes neither of them.

Only the innermost box produces data. Every claim you make is about the outermost one. The whole of this week is about making that leap defensible, and about saying plainly when it is not.

Example 1: three groups in one small study

You want the average weekly study time of students taking PS 361.

Population	All 420 students registered for PS 361 this semester.
Frame	The class register, which lists 395, because late registrations were never added.
Sample	The 180 who were in the lecture on the day you handed out the form and returned it.

Solution

The 25 missing from the register are a coverage problem; late registrants are plausibly the more disorganised students, who may study differently. The 215 on the register but not in your sample are non-response, and they are systematically the students who do not attend lectures, who are exactly the ones whose study time you would most want to know.

Note the direction: both gaps push your estimate of study time *upwards*. An estimate can be unbiased for the sample and badly wrong for the population.

3.2 Probability sampling

A sampling method is a *probability* method when every unit in the frame has a known, non-zero chance of selection. That property, not the size of the sample, is what licenses the inference.

3.2.1 Simple random sampling

Every subset of size n is equally likely. For a proportion p estimated from n units,

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}, \qquad 95\%\text{ margin of error} \approx 1.96 \times \mathrm{SE}(\hat{p}).$$

Where that formula comes from

It is worth seeing once, because the shape of it explains most of what follows. Write $X_i = 1$ if unit i has the characteristic and 0 otherwise. Then $\mathrm{E}(X_i) = p$ and

$$\mathrm{Var}(X_i) = \mathrm{E}(X_i^2) - [\mathrm{E}(X_i)]^2 = p - p^2 = p(1-p),$$

using $X_i^2 = X_i$ because the values are only 0 and 1. The estimator is $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$, so for independent draws

$$\mathrm{Var}(\hat{p}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\,p(1-p)}{n^2} = \frac{p(1-p)}{n},$$

and the standard error is its square root. Three consequences to carry with you:

- precision improves with $\sqrt{n}$, not n, so quadrupling the fieldwork halves the error and no more;
- $p(1-p)$ is largest at $p = 0.5$, which is why sample size calculations use that value when nothing is known;
- the population size is absent, because the draws were assumed independent. Section 3.2.5 puts it back when they are not.

Example 2: how precise is a sample of 200?

In a simple random sample of $n = 200$ students, 40% report working for pay during the semester. How precise is that?

Solution

$$\mathrm{SE} = \sqrt{\frac{0.40 \times 0.60}{200}} = \sqrt{0.0012} = 0.0346,$$

so the margin of error is $1.96 \times 0.0346 = 0.068$, about 6.8 percentage points. The honest sentence is: "40%, give or take about 7 points".

Notice what the formula does *not* contain: the size of the population. A sample of 200 is about as precise for 420 students as for all of Ghana, which surprises everyone the first time.

3.2.2 Systematic sampling

Take every kth unit from the frame after a random start. Easy in the field and usually as good as simple random sampling, with one trap: if the frame has a periodic structure with the same period as k, the sample locks onto it. Sampling every 7th day gives you the same weekday every time.

3.2.3 Stratified sampling

Divide the frame into groups (strata) that differ from each other, then sample within each. Done well it buys precision for free.

Split a population of N into H strata of sizes $N_1, \dots, N_H$, and write $W_h = N_h/N$ for the share of the population in stratum h. Draw n_h units from stratum h and let $\hat{p}_h$ be the proportion found there. Then

$$\hat{p}_{\text{str}} = \sum_{h=1}^{H} W_h \, \hat{p}_h, \qquad \text{SE}_{\text{str}} = \sqrt{\sum_{h=1}^{H} W_h^2 \, \frac{\hat{p}_h \, (1 - \hat{p}_h)}{n_h}} \, .$$

Under **proportional allocation**, $n_h = n \, W_h$: each stratum contributes in proportion to its size.

Notice what is *absent* from that standard error: any term for the differences *between* strata. Simple random sampling pays for that variation, because it leaves the mix of strata to luck; stratifying removes it, because every stratum is represented in exactly its right proportion. That is the whole of the gain.

Example 3: stratifying by programme

A cohort of 1200 students: 900 in engineering programmes, where 30% work for pay, and 300 in geomatics, where 70% do. You can afford $n = 200$.
With proportional allocation, $n_1 = 200 \times 900/1200 = 150$ from the first stratum and $n_2 = 50$ from the second. The overall proportion is

$$p = \frac{900(0.30) + 300(0.70)}{1200} = 0.40.$$

The stratified estimator's standard error is

$$\text{SE}_{\text{str}} = \sqrt{\left(\tfrac{900}{1200}\right)^2 \frac{0.3 \times 0.7}{150} + \left(\tfrac{300}{1200}\right)^2 \frac{0.7 \times 0.3}{50}} = 0.0324,$$

against $\text{SE}_{\text{SRS}} = \sqrt{0.4 \times 0.6/200} = 0.0346$.

Solution

The design effect is $(0.0324/0.0346)^2 = 0.875$: stratifying has done the work of a simple random sample about 14% larger, at no extra cost. The gain comes from removing between-programme variation from the error. Stratify on something that actually differs, and the gain is real; stratify on something irrelevant and you gain nothing, though you lose nothing either.

3.2.4 Cluster and multistage sampling

Sometimes you cannot reach individuals, only groups: you sample 20 halls and survey everyone in them, or 15 communities and visit every household. That is cluster sampling, and it is cheap, because travel dominates survey cost. It is also less precise, because people in the same cluster resemble one another.

The penalty is the **design effect**

$$\text{deff} = 1 + (m - 1)\,\rho,$$

where m is the number of units per cluster and ρ the intra-cluster correlation. Your *effective* sample size is $n_{\text{eff}} = n/\text{deff}$.
You rarely know ρ in advance. Take it from a pilot, from a previous round of the same survey, or from a published study of the same kind of cluster. For people grouped by where they live or study it is usually small but not negligible, around 0.01 to 0.10, and because deff depends on $m - 1$ times ρ, even a small value costs a great deal when the clusters are large.

Example 4: 600 students who count as 245

You survey 20 halls, 30 students in each, so $n = 600$. Students in the same hall share a water supply, a canteen and a study culture, giving $\rho = 0.05$.

Solution

$$\text{deff} = 1 + (30 - 1)(0.05) = 2.45, \qquad n_{\text{eff}} = \frac{600}{2.45} = 245.$$

Six hundred interviews bought the precision of 245. That is not an argument against cluster sampling, which may still be the only affordable design; it is an argument for saying so in the report, and for computing the margin of error from 245 rather than 600. Reporting ± based on 600 overstates your certainty by about 56%.
Note also that more clusters with fewer students each is almost always better: 10 students in each of 60 halls gives deff $= 1 + 9(0.05) = 1.45$.

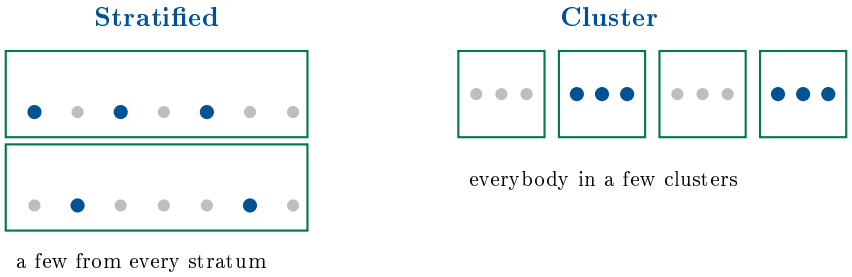


Figure 3.2: Filled circles are the people you interview. Stratifying spreads the sample across groups and buys precision; clustering concentrates it and buys cheap travel.

3.2.5 When the sample is a large share of the population

The independence assumption fails when you sample without replacement from a small population: each draw changes what is left. The correction is the *finite population correction* (FPC),

$$\text{SE}_{\text{FPC}}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \times \sqrt{\frac{N-n}{N-1}}.$$

When n is a small fraction of N the second factor is near 1 and is ignored. When it is not, it helps you a great deal.

Example 5: a class of 50, a sample of 40

You sample 40 of the 50 students in MF 281 and find half have a laptop.

Solution

Without the correction, $\text{SE} = \sqrt{0.25/40} = 0.0791$, a margin of about ± 15.5 points, which for a group you have nearly enumerated is absurd. With it,

$$\sqrt{\frac{50-40}{50-1}} = \sqrt{0.2041} = 0.4518, \qquad \text{SE}_{\text{FPC}} = 0.0791 \times 0.4518 = 0.0357,$$

a margin of about ± 7 points. If you had sampled all 50, the factor would be zero: a census has no sampling error at all, only measurement error.

3.2.6 Unequal selection probabilities and design weights

Sometimes you deliberately oversample a small group so that you can say something about it. The price is that your raw sample no longer resembles the population, and you must put it back together with weights.

If unit i had probability π_i of selection, its *design weight* is $w_i = 1/\pi_i$: the number of population members it stands for. The weighted estimator is

$$\hat{p}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}.$$

Example 6: the same study, ruined and then repaired by weights

Take the cohort from Example 4: 900 engineering students (30% work for pay) and 300 geomatics students (70% do). This time you sample 100 from each group, because you want a decent number of geomatics students.

Solution

The unweighted sample proportion is

$$\frac{100(0.30) + 100(0.70)}{200} = 0.50,$$

which is wrong: the true figure is 0.40. The sample contains far too many geomatics students, who work for pay more often.

The selection probabilities were $\pi = 100/900$ and $100/300$, so the weights are $w_1 = 9$ and $w_2 = 3$. Then

$$\hat{p}_w = \frac{9(100)(0.30) + 3(100)(0.70)}{9(100) + 3(100)} = \frac{270 + 210}{1200} = 0.40,$$

which is right. Each engineering respondent speaks for 9 students, each geomatics respondent for 3.

The habit

Whenever selection probabilities differ, by design or by accident, the analysis must carry weights. Software will happily compute an unweighted mean and tell you nothing is wrong.

3.3 Non-probability sampling

- **Convenience.** Whoever is in front of you. Fast, and the results generalise to nothing in particular.
- **Quota.** Fill fixed counts per group, choosing freely within. Looks like stratification, lacks the random selection that justifies the inference.
- **Snowball.** Participants recruit others. Sometimes the only way to reach a hidden population, and the sample inherits the shape of the social network.

These are not forbidden. They are honest when the write-up says what they are, and dishonest when a convenience sample is reported with a margin of error.

3.4 How many do I need?

For a proportion with margin of error e at 95% confidence, the worst case is $p = 0.5$, giving

$$n_0 = \frac{1.96^2 \times 0.25}{e^2}.$$

Example 5: sample size, with and without the population correction

You want a margin of error of 5 percentage points.

$$n_0 = \frac{(1.96)^2(0.25)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \approx 385.$$

If the population is large, 385 it is. But suppose the population is one cohort of $N = 1200$ students. Then the finite population correction applies:

$$n = \frac{n_0}{1 + \dfrac{n_0 - 1}{N}} = \frac{384.16}{1 + \dfrac{383.16}{1200}} = \frac{384.16}{1.3193} = 291.2 \approx 292.$$

Solution

Two lessons. Sampling a sizeable share of a small population is more efficient than people expect: 292 of 1200 does what 385 would in a city. And precision is expensive: tightening the margin from 5 to 3 percentage points needs $n_0 = 0.9604/0.0009 = 1068$, nearly three times the fieldwork for a modest gain.

3.5 Non-response, and how to be honest about it

Non-response matters only when the non-responders differ. The bias in the respondent mean is

$$\text{bias} = (1 - r)\,(\bar{y}_{\text{resp}} - \bar{y}_{\text{non}}),$$

where r is the response rate.

Example 6: a 60 percent response rate

A survey of study time achieves $r = 0.60$. Respondents average 1.8 hours per day. Suppose the non-responders truly average 3.2 hours.

Solution

The population mean is $0.60(1.8) + 0.40(3.2) = 1.08 + 1.28 = 2.36$ hours, so the reported figure understates it by 0.56 hours, about 24%. The formula gives the same: $(1 - 0.6)(1.8 - 3.2) = -0.56$.

Of course you do not know $\bar{y}_{\text{non}}$, or you would not have a problem. Two things you can still do. Show that respondents and the frame match on the characteristics you *do* know, such as programme and year. And report the bias that would follow from a plausible difference, as above: a reader can then judge whether your conclusion survives it. That is a sensitivity analysis, and it is worth more than a paragraph of apology.

3.5.1 A sensitivity table beats a disclaimer

You never know the non-responders' mean. What you can do is show how much it would have to matter. Holding the difference at 1.4 hours, the bias depends only on the response rate:

Response rate r	0.40	0.50	0.60	0.70	0.80	0.90
Bias (hours)	−0.84	−0.70	−0.56	−0.42	−0.28	−0.14
True mean (hours)	2.64	2.50	2.36	2.22	2.08	1.94

Read the table the other way and it becomes a design rule: at a 90% response rate even a large difference between responders and non-responders moves the answer by a tenth of an hour, while at 40% the study is mostly guesswork. Effort spent on response rate is effort spent on validity.

Chasing the response rate is usually better than modelling it. Four reminders to a class will do more for your study than any statistical correction, and cost nothing.

3.6 Observation or experiment?

	Observational	**Experimental**
Who decides who gets what	The world, or the participant	You, at random
Main threat	Confounding	Practicality and ethics
Supports	Association, description, prediction	Causal claims
Typical in your field	Maintenance logs, surveys, registers	Plant trials, A/B tests, laboratory work

Randomisation is the whole trick: it makes the groups comparable on everything, including the things you did not think to measure. Nothing else does that.

Example 9: randomisation does not promise perfect balance

Sixty students, of whom thirty do paid work, are split at random into two groups of thirty. How well balanced will the groups be on paid work?

Solution

The number of paid workers landing in the treatment group follows a hypergeometric distribution with mean 15. Summing its probabilities gives

$$P(13 \leq \text{paid workers in treatment} \leq 17) = 0.80.$$

So four times in five the split is within two students of even, and one time in five it is worse than that.

Two lessons that students often get backwards. Randomisation does not guarantee balance in any single experiment; it guarantees that imbalance is accidental rather than systematic, and that its effect is accounted for by the standard error. And with small groups, imbalance is common enough that it is worth blocking on the variables you know matter, which is simply stratification applied to an experiment.

3.6.1 Common observational designs

- **Cross-sectional.** One snapshot. Cheap, and cannot order cause and effect in time.
- **Cohort.** Follow a group forward. Expensive, but exposure precedes outcome.
- **Case-control.** Start from the outcome and look back. Efficient for rare outcomes, delicate about how controls are chosen.
- **Before and after.** The change is confounded with everything else that happened at the same time, unless you have a comparison group that did not change.

When somebody claims a change caused an improvement, ask what else changed that month. That question has sunk more confident conclusions than any statistical test.

3.7 Seven ways a sample misleads

1. **Frame gaps.** Whole groups absent from the list you sampled.
2. **Self-selection.** Those with strong feelings answer; the indifferent do not.
3. **Survivorship.** Only the units still present are observable.
4. **Periodicity.** Systematic sampling aligned with a cycle.
5. **Interviewer effects.** The answer depends on who asks, and how.
6. **Timing.** A survey run in the week before examinations measures the week before examinations.
7. **Ignoring the design.** Analysing a clustered or stratified sample as if it were simple random, which misstates every standard error.

3.8 Summary

Week 3 in six lines

- Population, frame, sample: name all three, and the gaps between them.
- Probability sampling is what licenses the inference; size alone does not.
- Stratify on what differs: precision for free.
- Clustering is cheap and costs precision; report the effective sample size.
- Quantify non-response bias with a plausible difference rather than apologising.
- Only randomisation makes groups comparable on what you did not measure.

3.9 Doing it on a computer

The arithmetic above is short enough to do by hand, which is why we did. In practice you will draw samples in software. Both snippets below draw a stratified sample of 200 from a frame and compute a design effect.

R

```
library(dplyr)
set.seed(42)

samp <- frame %>%                       # frame: one row per student
  group_by(programme) %>%               # the strata
  slice_sample(prop = 200 / nrow(frame))

deff  <- 1 + (m - 1) * icc              # m per cluster, icc = rho
n_eff <- nrow(samp) / deff
```

Python

```
import pandas as pd

samp = (frame
        .groupby("programme", group_keys=False)
```

```
        .apply(lambda g: g.sample(frac=200 / len(frame),
                                  random_state=42)))

deff  = 1 + (m - 1) * icc
n_eff = len(samp) / deff
```

Set the seed. Both snippets fix a random seed so that the same sample is drawn every time the script runs. Without it your results change on every run and nobody, including you, can reproduce the table in your report.

3.10 A checklist for any dataset you are handed

1. Who is the population, and who is on the list that was actually sampled?
2. How were units selected, and did everyone have a known chance?
3. What is the response rate, and who is missing?
4. Were selection probabilities equal? If not, where are the weights?
5. Is the sample clustered? If so, what is the effective sample size?
6. When was it collected, and what was unusual about that period?
7. Who paid for it, and what answer would have been convenient?

Seven questions, two minutes, and they will save you from most published nonsense.

3.11 Glossary

Coverage error	Population members with no chance of selection because they are not on the frame.
Design effect	Ratio of a design's variance to that of simple random sampling of the same size.
Design weight	$1/\pi_i$: how many population members a sampled unit stands for.
Effective sample size	n/deff: the simple random sample that would be as precise.
Estimand	The population quantity you are trying to learn (Week 2).
FPC	Finite population correction, $\sqrt{(N-n)/(N-1)}$.
Frame	The list from which the sample is actually drawn.
ICC (ρ)	Intra-cluster correlation: how alike units in the same cluster are.
Non-response bias	$(1-r)(\bar{y}_{\text{resp}} - \bar{y}_{\text{non}})$.
Probability sample	Every unit has a known, non-zero chance of selection.

Exercises

1. For a study of hostel water supply at UMaT, state a population, a realistic frame, and two groups that the frame would miss.
2. A simple random sample of 150 students finds 24% have never used the library. Give the standard error and the 95% margin of error.
3. Repeat Exercise 2 for $n = 600$. By what factor did the margin shrink, and by what factor did the fieldwork grow? What does that tell you about buying precision?

4. In Example 3, suppose both strata had 40% working for pay. Compute the stratified standard error and the design effect, and explain the result in one sentence.

5. A survey takes 25 students from each of 12 halls, with $\rho = 0.08$. Compute the design effect and the effective sample size. How many halls would you need, at 10 students each, to reach the same effective size?

6. You need a margin of error of 4 percentage points in a cohort of 800. Compute n_0 and the finite-population-corrected n.

7. A clinic survey achieves a 45% response rate. Respondents report a mean waiting time of 32 minutes. If non-responders truly waited 50 minutes on average, what is the true mean and the bias?

8. For Exercise 7, how large would the response rate have to be for the bias to fall below 5 minutes, assuming the same difference between the groups?

9. Classify each as observational or experimental, and name the main threat: (a) comparing failure rates of two pump brands already installed; (b) randomly assigning half the hostels to a new filter; (c) comparing examination results before and after a syllabus change.

10. A manager samples every 10th truck leaving the pit for a load check. Describe a plausible periodic structure that would make this misleading.

11. The class wellbeing survey is open to anyone who chooses to fill it in. Name its population, its frame and the selection problem, and suggest one change that would improve it without identifying anybody.

12. Design, in half a page, a sample for estimating the proportion of UMaT students who do paid work during term. State the design, the sample size, the expected margin of error, and the two biggest threats to validity.

Answers to the odd-numbered exercises

1. Population: every student living in UMaT hostels this semester. Frame: the accommodation office's allocation list. Missed: students who swapped rooms informally, and students living in private hostels who use campus water.

3. $\text{SE} = \sqrt{0.24(0.76)/600} = 0.0174$, margin 0.0342, about 3.4 points, against 6.8 points at $n = 150$. The margin halved while the fieldwork quadrupled: precision costs the square of what you want.

5. $\text{deff} = 1 + 24(0.08) = 2.92$, so $n_{\text{eff}} = 300/2.92 = 103$. With 10 students per hall, $\text{deff} = 1 + 9(0.08) = 1.72$, so reaching 103 needs $103 \times 1.72/10 \approx 18$ halls. Fewer people, more places, same precision, more travel.

7. True mean $= 0.45(32) + 0.55(50) = 14.4 + 27.5 = 41.9$ minutes. The reported 32 minutes understates it by 9.9 minutes.

9. (a) Observational; the brands are installed in different places, so location confounds the comparison. (b) Experimental; the threat is practical, since hostels talk to each other and the filters may be shared. (c) Observational and before-after; everything else that changed that year is confounded with the syllabus.

11. Population: students in the two courses. Frame: nobody wrote one, so in effect those who visit the course page. Problem: self-selection, since students who are coping least well may be least likely to answer. An improvement that identifies nobody: hand out the link in class and give everyone five minutes to complete it there, which converts a self-selected sample into something much closer to the whole class.

Before next week

Read *OpenIntro Statistics* Sections 1.3 to 1.5. Week 4 turns to what you do once the data is in front of you: describing a single variable honestly, and noticing when an average is hiding more than it shows.